Benjamin Diedrichsen

Fullstack Software Engineer

github: https://github.com/bennidi
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The design and construction of software systems - from core libraries, internal DSLs, service APIs to mid-size software landscapes - is my job and one of my passions.

This includes many aspects of the software development lifecycle: requirements engineering, feature planning, data and domain modeling, prototyping, deployment, production.

Polyglot

I write code in various languages using combinations of object oriented and functional paradigm. @Home on the JVM [Java, Scala, Groovy], I recently became an enthusiastic user of the JavaScript ecosystem - node.js, coffeescript, livescript, react.js

Everything is a model

I enjoy the creative task of deconstructing reality into appropriate models of objects, functions, data structures and information flows... and I love to write beautiful and efficient code.

I learned that well-factored code and simple but solid solutions often win over bleeding edge technology. And some technologies clearly win over others...

I also believe in the values of agile methodologies: open and honest communication, collective learning and team collaboration are as important as choosing the right technology stack.

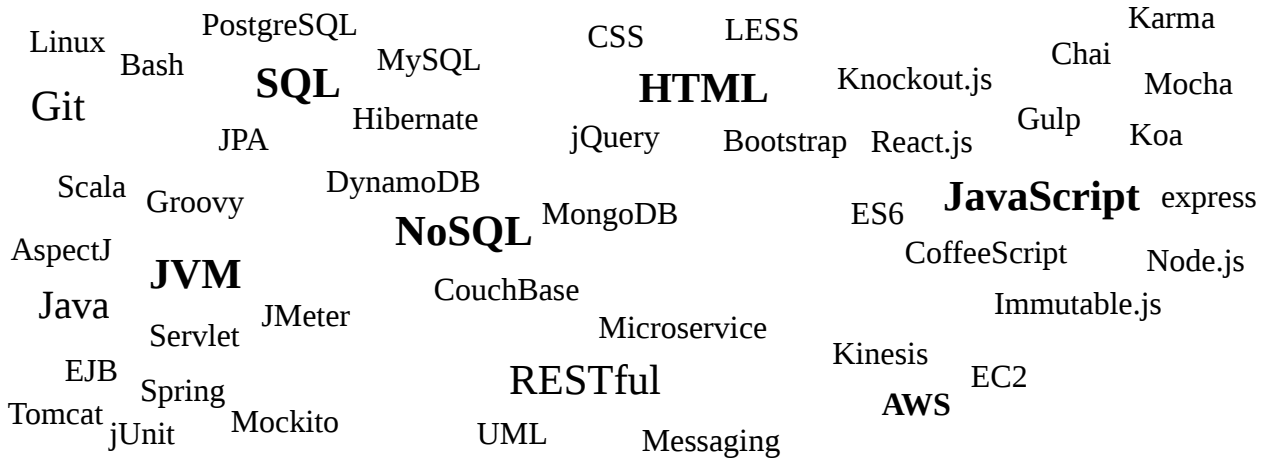
You can hire me to

- design and prototype core modules or software architecture
- fix performance issues in application code or data models
- plan and implement refactorings
- do code reviews and help improve code quality
- support the evaluation of technological decisions

I am especially interested if..

- you solve social problems instead of consumer problems
- you have a sound technology stack
- you have an awesome team and fun at the workplace

When I am not working on codebases, I go climbing, play the guitar, learn about bodywork, psychotherapy or permaculture.



Software projects

12/2016 – 07/2017

Fullstack Software Engineer (Java, Spring Boot, Node.js)

Affinitas GmbH

eDarling, ElitePartner

Development of different backend services in a software landscape powering multiple online platforms in eDating sector. The services were mainly Java&Spring applications. A transition to a more microservice style architecture with automated AB-deployment on AWS and service discovery with Eureka was onoing. Work was organized in agile teams following SCRUM methodology.

Technologies

Java 8, Spring Boot, Gradle, SQS, DynamoDB, Kinesis, Node.js, CoffeeScript, koa, gulp, RAML, JWT, Eureka, Spring Config, Jenkins, Spinnaker, RESTful

Tasks

- Development of RESTful API for sending push notifications to mobile clients. The service was built with Spring Boot and Java. Instance local message queue with automatic overflow into and import from SQS
- Prototyping of Node.js microservice template. Integration with Eureka and Spring Config
- Design of RESTful API for in-app micropayment to be consumed by mobile clients (iOs, Android). API first approach with RAML. Implemention on Node.js stack with CoffeeScript and koa middlewares. Integration with existing service landscape (billing, reporting, legacy app backend) using service to service calls

05/2016 – 10/2016

Frontend Engineer (CoffeeScript, React)

www.mein-schulplan.de

Rewrite of existing web application to new technology stack and mobile first approach. The old angular app lacked unit testing and the persistence model frequently produced errors due to lost updates. Performance was an issue. The new app should be shipped with unit and e2e tests, run smoothly on mobile devices and be tolerant to connection loss (offline first approach).

Technologies React, CoffeeScript, ES6, Gulp, PouchDB, jQuery, Bootstrap, LESS

Tasks

- Technology evaluation and implementation of first prototype with React and CoffeeScript
- Design and prototyping of immutable data model for lecture plans and class schedule
- Implementation of algorithm to derive lesson plans from lecture plans and map to class schedule
- Implementation of local storage with auto-syncing capabilities to backend storage
- Integration with jQuery calendar plugin

07/2015 – 12/2015 **Backend Developer (Scala)**

Okotta GmbH
www.okotta.com

Development of backend services and back office tools for a mobile app (iOS, Android) in insurance market. The team's mission was to build a relatively complex MVP. The underlying domain model and insurance data was complex and a moving target. A flexible RESTful API with good query capabilities was built with with JVM technologies like Scala, Java 8, Spring Boot, Gradle. Standalone API integration tests written in Groovy were run during CI pipeline (Jenkins).

Technologies Scala 2.11, Scalatra, ScalDI, reactivemongo, MongoDB, JSON Web Token, Swagger, WireMock, React, ES6

Tasks

- Requirements engineering and modeling of domain classes
- Prototyping of RESTful API for core domain with Scala and Scalatra
- Evaluation of persistence technologies in Scala stack: Slick, MongoDB, Couchbase
- Implementation of various RESTful data repositories
- Integration of WireMock for fullstack API tests using JSON fixtures
- Prototyping of back-office tool with React

06/2014-12/2014 **Backend Developer**

Small Improvements
small-improvements.com

Java Web application running on GAE (servlet like environment). BigTable was one and only storage model and bad fit for existing and relation heavy domain model. Extensive use of Memcache. Wicket and Angular were used as frontend technologies. A migration towards a more RESTful backend and decoupled frontend was ongoing.

Technologies Google App Engine, Java 6, Wicket, BigTable, Memcache, JMeter, JProfiler, Jackson, Groovy, Spock, Gradle

Tasks

- (Re-)Implementation of REST-API layer with composable serializers and less boilerplate code
- Performance optimization of object serialization based on JMeter and JProfiler insights
- Integration of caching layer using Guava Cache
- Refactoring to patterns

03/2013 - 09/2013	**Senior Software Engineer (Java 7, J2EE, EJB)**
ProfitBricks GmbH *www.profitbricks.com*	J2EE backend services for calculation of virtual data center components and placement onto high-performance physical hardware (provisioning). Frontend GUI as well as python client make provisioning requests to backend which produced an intermediate message format to be consumed by external programs that manage physical resources in the data centers.
Technologies	J2EE, Java 7, EJB 3.1, EclipseLink, JPA 2.1, Glassfish, jUnit, cglib, OSGI, Postgress, VisualVM, H2, Git, Maven, Jenkins
Tasks	➢ Profiling and performance optimizations of production code ➢ Implementation of start/stop feature that allows virtual servers to be stopped and restarted manually ➢ Rewrite of test code with internal DSL and H2 and lazy evaluation using cglib proxies ➢ Code reviews and refactorings
02/2012 - 12/2012	**Fullstack Software Engineer (Java, Spring, JavaScript)**
Alecto GmbH *www.friendsurance.de*	Java Spring backed online platform offering a social network like environment for insurance customers. Very traditional three-tier application with Spring Services, Spring Web MVC and persistence layer built on top of Hibernate.
Technologies	Java 7, EJB 3.1, Hibernate (JPA 2.1), Spring 3, Tomcat 7, Glassfish, jUnit, JSF 2, Selenium, jQuery, Postgres, Servlet, Git, Maven
Tasks	➢ Migration from EJB 3.1 (Glassfish) to Spring (Tomcat) ➢ Redesign of the persistence abstraction layer ➢ Fully historized social graph based on immutable data ➢ Caching layer with synchronization across machines ➢ Built selenium test infrastructure on grid server

Co-operative course of studies

15.10.2007 – 30.09.2008	**Softwaredevelopment ottogroup retail systems**
www.ottogroup.com	Preparation and partial migration of the object relational mapping layer (JDO → JPA) in a large J2EE landscape. 100+ domain classes with multiple client systems reading from services.
Technologies	Java 6, JDO 2.0, JPA 1.0, jUnit, Spring, Xtext, Xtend,Xpand,
Tasks	➢ Extension of existing persistence infrastructure for operating JDO and JPA in parallel (soft migration strategy) ➢ Design and implementation of small model-to-model transformation language using Xtext, Xtend, Xpand ➢ Implementation of mapping rules to transform existing JDO mappings

Degree program

01.10.2004 –30.09.2008	**Co-operative degree program 'Economics and computer science (Dipl. FH)' in co-operation with ottogroup**
Focue areas	Model driven software development ■ Modern software technologies ■ Datastream Managementsystems
Diploma thesis	Implementation of a generic transformation component for XML based models and its application to a model migration problem
Final grades	1.0 (thesis), 1.5 (diploma)

Open Source Projects

[github:mbassador]	High-Performance Java Event Bus library: Implementation of publish-subscribe pattern. Low memory footprint, nonblocking iterators, high throughput
[github:mailody]	Online trash mail service built with Node.js, MongoDB, restify, knockout.js, HTML 5, Java client for Selenium
[github:kermit]	Kermit is an elegant, stream-based, extensible web scraper system. It is packed with pluggable features and offers a plugin mechanism for easy extension. Written in Node.js and coffeescript with NeDB as datastore.

Languages

*Foreign**

	oral	writing
English	C1	C2
Spanish	B2	C1
French	A1	A2

** Self evaluation according to refernce framework of European Language Portfolio (see appendix or www.coe.int/portfolio)*

Native Deutsch